

2.5 Maxwell's Equations

Electromagnetic behavior can be described using a set of four fundamental relations known as *Maxwell's Equations*. These equations are a mathematical model for observations about electric and magnetic fields discovered by Faraday, Ampère, and others. Maxwell's equations in point form are

$$\nabla \times \bar{E} = -\frac{\partial \bar{B}}{\partial t} \quad (2.35)$$

$$\nabla \times \bar{H} = \bar{J} + \frac{\partial \bar{D}}{\partial t} \quad (2.36)$$

$$\nabla \cdot \bar{D} = \rho_v \quad (2.37)$$

$$\nabla \cdot \bar{B} = 0 \quad (2.38)$$

The first equation is known as Faraday's law, the second is Ampère's law, and the third and fourth are Gauss's laws for the electric and magnetic fields. The field and source quantities are

$\bar{E}$: Electric field intensity	V/m
$\bar{H}$: Magnetic field intensity	A/m
$\bar{D}$: Electric flux density	C/m ²
$\bar{B}$: Magnetic flux density	Wb/m ²
$\bar{J}$: Electric current density	A/m ²
ρ_v : Electric charge density	C/m ³

The first four quantities represent the electromagnetic field. The last two quantities are sources that radiate electromagnetic fields.

In free space (i.e., a vacuum), the electric and magnetic field intensities and flux densities are related by constitutive relations, which are

$$\bar{D} = \epsilon_0 \bar{E} \quad (2.39a)$$

$$\bar{B} = \mu_0 \bar{H} \quad (2.39b)$$

where $\epsilon_0 \simeq 8.854 \times 10^{-12}$ F/m is the permittivity of free space and $\mu_0 = 4\pi \times 10^{-7}$ H/m is the permeability. Air affects the permittivity slightly, but we generally approximate the permittivity of air by the free space value. The effect of dielectrics and magnetic materials on the electromagnetic field can be modeled as a change in the value of ϵ and μ .

Maxwell's equations can also be expressed in integral form:

$$\oint_C \bar{E} \cdot d\bar{\ell} = -\frac{d}{dt} \int_A \bar{B} \cdot d\bar{s} \quad (2.40)$$

$$\oint_C \bar{H} \cdot d\bar{\ell} = \int_A \bar{J} \cdot d\bar{s} + \frac{d}{dt} \int_A \bar{D} \cdot d\bar{s} \quad (2.41)$$

$$\oint_S \bar{D} \cdot d\bar{s} = \int_V \rho_v dV = Q \quad (2.42)$$

$$\oint_S \bar{B} \cdot d\bar{s} = 0 \quad (2.43)$$

where S is the closed surface bounding the volume V and C is the closed path bounding the area A . We will show how the integral and point forms of these equations are related a little later.